Scoreboard:

$$\frac{24}{9.55k} = 2.51 \text{ V/krpm}$$

$$9.5493 \times 10^{-3} k_e = .024$$

Notes ch 22 and 23.2-23.3

back emf: $E = k_e w$ $\quad k_e$ = speed or voltage const

$w$ = rad/s $\quad T_{motor} = T_{load} + T_{friction}$

$k_e$ = V/(rad/s)

$E$ = volts

$$\text{power (watts)} = E(V) * I(A) = Tw = k_e w I$$

$T = k_e I$ $\quad k_T$ = Nm/A torque const

$$k_T \text{ (in·oz/A)} = 1.3542 * k_e \text{ (V/krpm)}$$

$$k_T \text{ (Nm/A)} = 9.5493 * 10^{-3} k_e \text{ (V/krpm)}$$

at stall, $w = 0$ $\quad k_T \text{ (Nm/A)} = k_e \text{ (V/(rad/s))}$

$$V = \frac{T}{k_T} R + k_e w \qquad T_{stall} = \frac{w_{no\ load}}{R_m}$$

← speed reg const

$$R_m = \frac{R}{k_T k_T}$$

$$P = \frac{VT}{k_e} - R_m T^2$$

$$P = Tw = (T_f - T_L) w \qquad P_{max} = \left(\frac{k_T}{4 k_e R}\right) V^2$$

eff: $\eta = \frac{P_{out}}{P_{in}} = \frac{(T_m - T_f) w}{VI}$

$$\eta_{max} = \left(1 - \sqrt{\frac{T_f R}{K_T V}}\right)^2$$

Gear ratio $N = \frac{\omega_{in}}{\omega_{out}}$

$$T_{out} = \eta T_{in} N$$

23: (essentially EIEN 50 and 100)

$$P = \frac{1}{2} I^2_{stall} L \times freq$$

Problem 1:

1. DC motor w/ $K_T = 2.0$ Nm/A, 10 V, at stall, .2 Ω stall torque?

$$V = \frac{T}{K_T} R + K_e \omega \rightarrow \omega = 0 \rightarrow V = \frac{T}{K_T} R$$

$$10 = \frac{T}{2}(.2) \rightarrow \frac{10(2)}{.2} = T = \boxed{100 \text{ nM}}$$

2. same motor at 8V, if T=60, what is ω (in rad/s)

$$8 = \frac{60}{2}(.2) + K_e \omega \rightarrow K_e = K_T \text{ b/c of units} \rightarrow 8 = \frac{60}{2}(.2) + 2\omega$$

$$8 = 30(.2) + 2\omega \rightarrow 2 = 2\omega \rightarrow \boxed{\omega = 1 \text{ rad/s}}$$

3. Power prod by motor in Q2?

$\frac{60}{2} = 30A \rightarrow 8V(30A) = 240W$ consumed but not produced

$P_{prod} = T\omega = 60(1) = \boxed{60W}$ bad eff.

4. efficiency of motor at 28.65 RPM w/ 10 Nm load

$$28.65\left(\frac{2\pi}{60}\right) = 3.000 \text{ rad/s}$$

$$V = \frac{10}{2}(.2) + 2(3) = 7 \rightarrow 7\left(\frac{10}{2}\right) = 35V \text{ pm}$$

$$T\omega = P_{out} = 10(3) = 30W$$

$$\eta = \frac{30}{35} = \boxed{85.71400} \text{ not bad}$$